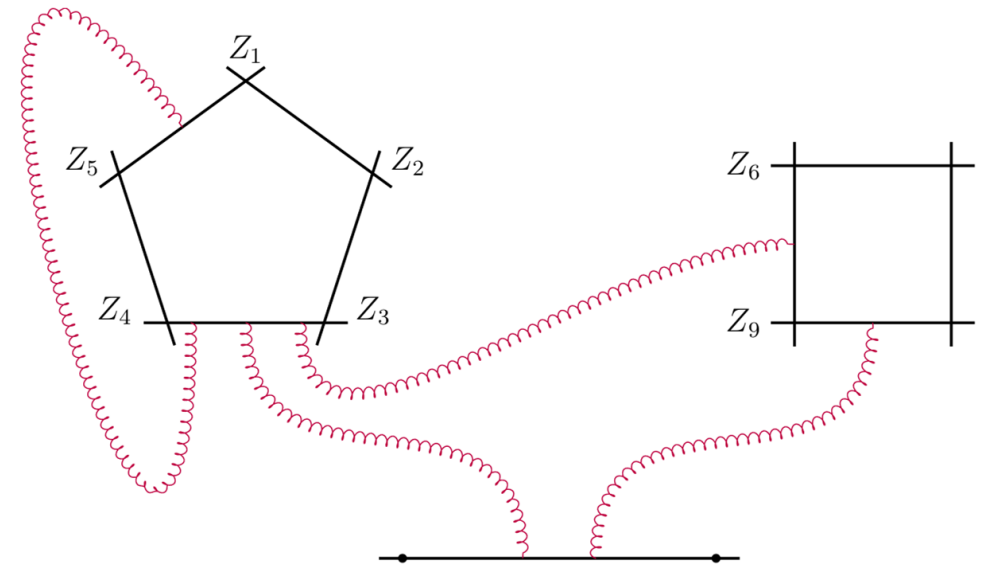


Light-like Wilson loop correlators in $\mathcal{N} = 4$ SYM

BASED ON 2512.12655, 2601.23210, 2602.18423

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Overview

- Light-like Wilson loop correlators in space-time
- Super-Wilson loop correlators in twistor space
- Tree-level and 1-loop correlators
- Chiral box expansion and factorisation
- $\bar{Q}$ equation
- Wilson loop recursion relation
- Ongoing work and future directions

Light-like Wilson loop

We consider $\mathcal{N} = 4$ SYM with either $U(N)$ or $SU(N)$ gauge group. A Wilson loop operator is defined as,

$$\mathcal{L}(C) = \frac{1}{N} \text{tr} \mathcal{P} \exp i \oint_C dx^\mu A_\mu$$

where we take C to be a closed piecewise light-like contour. The expectation value has the following form,

$$W_n = \langle \mathcal{L}(C) \rangle = \left[\prod_{i=1}^n D_i \right] F_n R_n$$

D_i is the divergent part, F_n is a chosen finite part and R_n is finite and conformally invariant. We are interested in the expression at weak coupling.

$$g^2 \equiv \frac{g_{\text{YM}}^2 N}{16\pi^2}$$

These are well studied objects due to the duality between Wilson loops and scattering amplitudes.

Wilson loop correlators

Correlation functions of multiple Wilson loops have a similar form to single Wilson loops [Drummond, Gürdoğan, MR, Wright, 2025],

$$W_{n_1, \dots, n_m} = \langle \mathcal{L}(C_1) \dots \mathcal{L}(C_m) \rangle = \prod_{r=1}^m \left[\left[\prod_{i=1}^{n_r} D_{i_r} \right] F_{n_r} \right] R_{n_1, \dots, n_m}$$

We are interested in these observables as they may tell us more about correlators of Wilson loops and Lagrangians, important observables in the study of amplitudes and may tell us about the spectrum of operators in $\mathcal{N} = 4$ SYM.

$W_{n_1, \dots, n_r}$ can be separated into connected and disconnected parts.

$$W_{n_1, n_2} = W_{n_1} W_{n_2} + W_{n_1, n_2}^{\text{conn}}$$

Due to the factorisation of the correlator, the finite, conformally invariant part can be expressed as

$$R_{n_1, n_2} = R_{n_1} R_{n_2} + R_{n_1, n_2}^{\text{conn}}$$

Super-Wilson loop correlators

At $O(g^2)$, the Feynman diagrams are of the form



We can consider the extension of multiple Wilson loops to super Wilson loops. Although these objects are no longer dual to amplitudes, we use $N^k MHV$ to refer to the Grassmann degree $4k$ part of the correlator.

$$\mathcal{W}_{n_1, \dots, n_m} = \sum_k \mathcal{W}_{n_1, \dots, n_m}^{(k)} \quad \mathcal{W}_{n_1, \dots, n_m} = \left[\prod_r \left[\prod_{i=1}^n D_i \right] F_{n_r} \right] \mathcal{R}_{n_1, \dots, n_m} \quad \mathcal{R}_{n_1, \dots, n_m}^{(k), \text{conn}} = \sum_l g^{2l} \mathcal{R}_{n_1, \dots, n_m}^{(k, l), \text{conn}}$$

Super Wilson loop correlators depend on the supertwistors $\mathcal{Z}_{r, i_r} = (Z_{r, i_r} | \chi_{r, i_r})$, $r = 1, \dots, m$, associated to the cusps of the Wilson loops, with Z_{r, i_r} and χ_{r, i_r} are bosonic and fermionic variables respectively.

Correlators in twistor space

Perturbatively, maximally supersymmetric Yang-Mills theory is given in twistor space by holomorphic Chern-Simons theory on $CP^{3|4}$ and an extra piece which captures interaction terms [Boels, Mason, Skinner, 2007].

$$S(\mathcal{A}) = S_1(\mathcal{A}) + S_2(\mathcal{A})$$

$$S_1(\mathcal{A}) = \frac{iN}{8\pi^3} \int D^{3|4} \mathcal{Z} \wedge \text{tr} \left(\mathcal{A} \wedge \bar{\partial} \mathcal{A} + \frac{2}{3} \mathcal{A} \wedge \mathcal{A} \wedge \mathcal{A} \right) \quad S_2(\mathcal{A}) = \frac{g^2 N}{\pi^2} \int d^{4|8} X \log \det (\bar{\partial} + \mathcal{A})_X$$

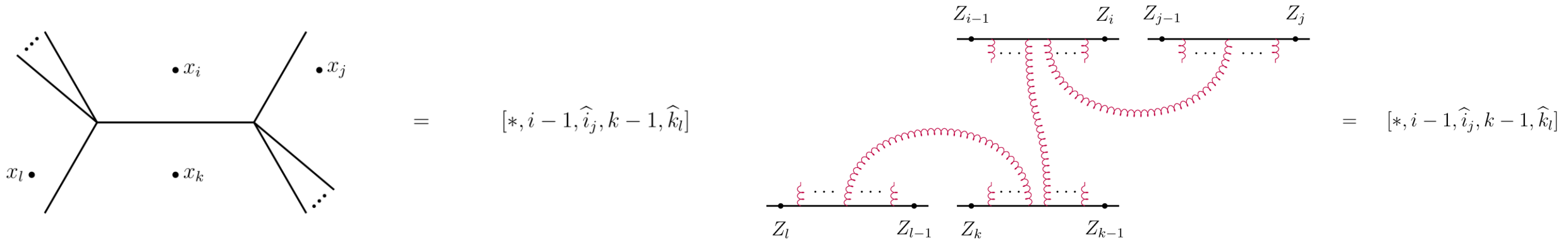
$$\begin{aligned} \mathcal{A}(Z, \chi) = & a(Z, \bar{Z}) + \chi^{A'} \tilde{\gamma}_{A'}(Z, \bar{Z}) + \frac{1}{2!} \chi^{A'} \chi^{B'} \phi_{A'B'}(Z, \bar{Z}) \\ & + \frac{1}{3!} \epsilon_{A'B'C'D'} \chi^{A'} \chi^{B'} \chi^{C'} \gamma^{D'}(Z, \bar{Z}) + \frac{1}{4!} \chi^1 \chi^2 \chi^3 \chi^4 g(Z, \bar{Z}) \end{aligned}$$

In twistor space, the Wilson loop is defined as [Mason, Skinner, 2010],

$$\mathcal{L}(C) = \frac{1}{N} \text{tr} \mathcal{P} \prod_{i=1}^n \sum_{l_i=0}^{\infty} (-\bar{\partial}_i^{-1} \mathcal{A}(Z_i(0)))^{l_i} \quad (\bar{\partial}_X^{-1} \omega)(s) = \int_X G(s, s') \wedge \omega(s') \quad G(s, s') = -\frac{1}{2\pi i} \frac{ds'}{(s - s')}$$

R -invariant form of Feynman rules

According to the Wilson loop/scattering amplitude duality, the MHV diagrams of amplitudes are in a 1-1 correspondence with the twistor diagrams of the dual Wilson loop [Mason, Skinner, 2010]. The rule for a propagator in an MHV diagram is equal to the rule for a propagator in the dual Wilson loop diagram.



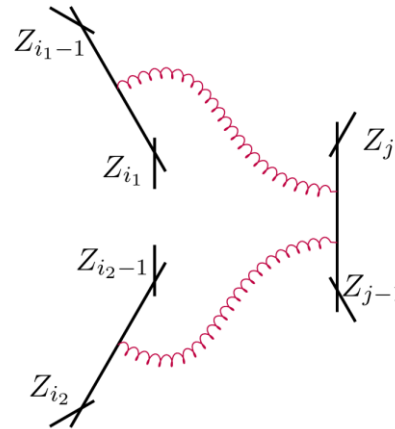
$$[a, b, c, d, e] = \frac{\delta^{0|4}(\chi_a \langle bcde \rangle + \text{cyc.})}{\langle abcd \rangle \langle bcde \rangle \langle cdea \rangle \langle deab \rangle \langle eabc \rangle}$$

$$\langle ijkl \rangle = \det(Z_i Z_j Z_k Z_l)$$

where $\hat{i}_j = (i - 1 \ i) \cap (* j - 1 \ j)$. If there is no propagator attached to the same line in the clockwise direction, then $\hat{i}_j$ is replaced by i .

Tree-level correlators

If we consider the connected part of an NMHV correlator (Grassmann degree 4), we sum over all ways of attaching one propagator between the two Wilson loops as the propagator has Grassmann degree 4. For N^2MHV , we sum over all ways of attaching two propagators between the different sides of the Wilson loops, and so on.



Tree-level correlators

The $NMHV$ and N^2MHV results have a very simple form,

$$\mathcal{W}_{n_r, n_s}^{(1,0), \text{conn}} = \frac{1 - \alpha}{N^2} \sum_{i,j} [* , i - 1, i, j - 1, j]$$

$$\mathcal{W}_{n_1, n_2, (a)}^{(2,0), \text{conn}} = \frac{1}{2N^4} (N^2 - 1) \left(\sum_{i,j} [* , i - 1, i, j - 1, j] \right)^2$$

where the second expression is obtained from combining diagrams. $\alpha = 1$ for $SU(N)$ and $\alpha = 0$ for $U(N)$ and second expression is for $SU(N)$.

Loop-level correlators

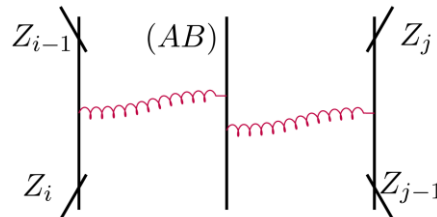
Taking the path integral form of the expectation value of a single Wilson loop,

$$\langle \mathcal{L}(C) \rangle = \int [d\mathcal{A}] e^{-(S_1 + S_2)} \mathcal{L}(C)$$

We see that the $O(g^2)$ part of the correlator depends on the correlator of a Wilson loop and another extended operator,

$$\begin{aligned} \langle \mathcal{L}(C) \rangle &= \int [d\mathcal{A}] e^{-S_1} \mathcal{L}(C) \left[1 + \frac{g^2 N}{\pi^2} \int d^4|8 X \sum_{r=2}^{\infty} \frac{1}{r} \text{tr}(-\bar{\partial}_X^{-1} \mathcal{A})^r + O(g^4) \right] \\ &= \langle \mathcal{L}(C) \rangle^{\text{CS}} + \frac{g^2 N}{\pi^2} \sum_{r=2}^{\infty} \frac{1}{r} \int d^4|8 X \langle \mathcal{L}(C) \text{tr}(-\bar{\partial}_X^{-1} \mathcal{A})^r \rangle^{\text{CS}} + O(g^4). \end{aligned}$$

To compute the integrand of the $O(g^2)$ part of the correlator, we sum over diagrams with propagators attached between the Wilson loops and the extra extended operator which lies on the line $X = (AB)$.



$O(g^2)$ correlators

Using the Feynman rules, the *MHV* 1-loop diagrams, we get

$$\mathcal{W}_{n_1, n_2}^{(0,1), \text{conn}} = \frac{1}{N^2} \sum_{i,j} K_{ij}$$

$$K_{ij} = \int \frac{d^{4|8} x_{AB}}{\pi^2} [* , i-1, i, A, B'] [* , j-1, j, A, B'']$$

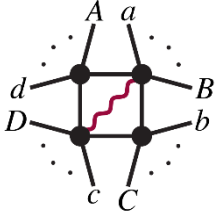
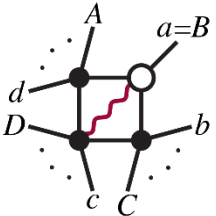
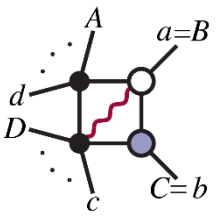
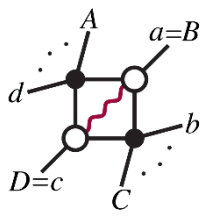
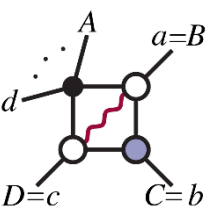
Performing the fermionic integration, taking the total derivative and noting the cancellation amongst integrals, we find

$$\mathcal{W}_{n_1, n_2}^{(0,1), \text{conn}} = \frac{1}{N^2} f_{n_1, n_2}$$

$$f_{n_1, n_2} = \sum_{i,j}^{n_1, n_2} \text{Li}_2(1 - v_{ij}) + \sum_{\substack{k \leq i \\ j \leq l}} \log v_{ij} \log v_{kl} \quad u_{i,j,k,l} = \frac{x_{ij}^2 x_{kl}^2}{x_{il}^2 x_{kj}^2}, \quad v_{ij} = u_{i,j,i+1,j+1}$$

Chiral box expansion and factorisation

Amplitudes can be expressed in terms of simple integrands/integrals called *chiral boxes* [Bourjaily, Caron-Huot, Trnka, 2013], which capture the singularities of the amplitude.

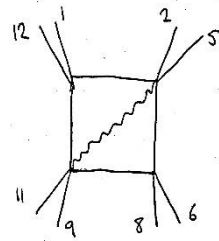
4-mass	3-mass	2-mass easy	2-mass hard	1 mass
				

The cut associated to the poles of each chiral box of the 1-loop integrand gives a **product of tree-level amplitudes**.

Thus, the amplitude can be expressed in terms of a small set of simple integrals with coefficients given by tree-level amplitudes.

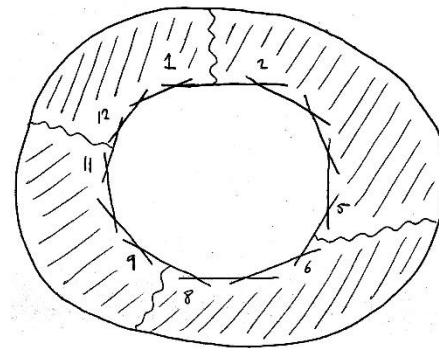
Chiral box expansion and factorisation

The tree-level amplitude coefficients can be understood from the partitioning of the 1-loop Wilson loop Feynman diagrams [Drummond, MR, Wright, 26]. Consider a 1-loop dodecagon at $N^3\text{MHV}$ and the 4m chiral box within the box expansion:



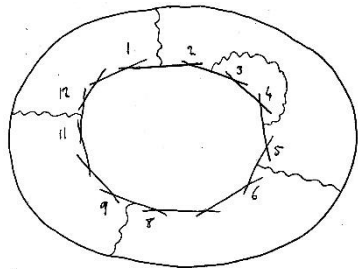
$$\sim \frac{1}{\langle AB12 \rangle \langle AB56 \rangle \langle AB89 \rangle \langle AB1112 \rangle}$$

For a Feynman diagram to contribute this cut, it must have 4 propagators from the Lagrangian to (12), (56), (89), (11,12). 4 regions are formed.



Chiral box expansion and factorisation

The 5th propagator must stay in the same region (planarity). Consider the following diagram:

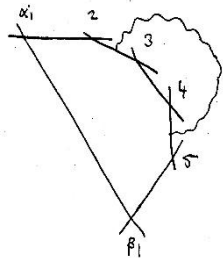


$$= [* , A, (B)_6, 1, 2][* , A, (B)_9, 5, 6][* , A, (B)_{12}, 8, 9][* , A, (B)_2, 11, 12] \\ \times [* , 2, 3, 4, 5]$$

Taking the cut of this diagram, the 4 Lagrangian propagators give a Grassmann factor:

$$[\delta_1, 1, 2, 5, 6][\beta_1, 8, 9, 11, 12]$$

5th factor is unchanged. This is a diagram in $\langle W(\alpha_1, 2, 3, 4, 5, \beta_1) \rangle^{tree}$.

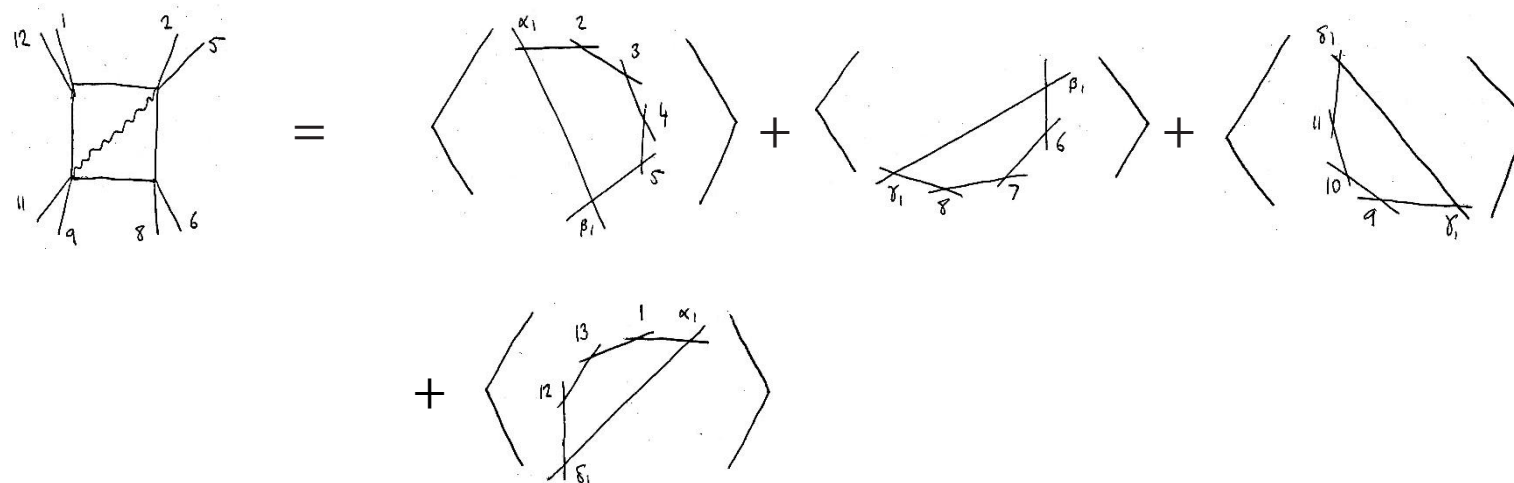


$$= [* , 2, 3, 4, 5]$$

Chiral box expansion and factorisation

The sum over the cuts of diagrams with the extra propagator in this region is equal to $[\alpha_1, 1, 2, 3, 4][\beta_1, 5, 6, 7, 8]\langle W(\alpha_1, 2, 3, 4, 5, \beta_1) \rangle^{tree}$. Similarly, the sum over cuts of diagrams with the extra propagator in the other regions are equal to tree-level Wilson loops formed by these other regions.

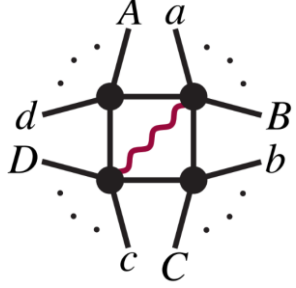
Coefficient of



$$= \langle \text{Diagram 1} \rangle + \langle \text{Diagram 2} \rangle + \langle \text{Diagram 3} \rangle + \langle \text{Diagram 4} \rangle$$

At higher MHV degree, it can be seen that this is factorisation into products of tree-level Wilson loops.

Chiral box expansion and factorisation

Coefficient of  =
$$\varphi_1[\delta_1 A a B b][\beta_1 C c D d] \times$$
$$\mathcal{A}_a(\alpha_1, \dots, \beta_1) \mathcal{A}_b(\beta_1, \dots, \gamma_1)$$
$$\mathcal{A}_c(\gamma_1, \dots, \delta_1) \mathcal{A}_d(\delta_1, \dots, \alpha_1)$$

[Bourjaily, Caron-Huot, Trnka 2013]

$$\ell_1 \left\{ \begin{array}{l} \alpha_1 = (aA) \cap (Bb \gamma_1) \\ \beta_1 = (Bb) \cap (aA \delta_1) \\ \gamma_1 = (cC) \cap (Dd \alpha_1) \\ \delta_1 = (Dd) \cap (cC \beta_1) \end{array} \right.$$

For multiple Wilson loops, the story is very similar, except there are more types of chiral boxes and types of regions [Drummond, MR, Wright, 2026]. With these extra chiral boxes and the derivation of chiral box coefficients, we have the integrated expression for any 1-loop correlator!

$\bar{Q}$ equation

Not all integrals are known for basis integrands at higher loops, so we must look to alternative methods for finding integrated loop amplitudes. One such way is by using the $\bar{Q}$ equation. A generator of a part of dual superconformal symmetry is,

$$\bar{Q}_a^A = \sum_{i=1}^n \chi_i^A \frac{\partial}{\partial Z_i^a}$$

It has the following action on the Wilson loop [Caron-Huot, He, 2014]:

$$\bar{Q}_a^A R_{n,k} = a \operatorname{Res}_{\epsilon=0} \int_{\tau=0}^{\tau=\infty} \left(d^{2|3} \mathcal{Z}_{n+1} \right)_a^A \left[R_{n+1,k+1} - R_{n,k} R_{n+1,1}^{\text{tree}} \right] + \text{cyclic}$$

$$a := \frac{1}{4} \Gamma_{\text{cusp}} = g^2 - \frac{\pi^2}{3} g^4 + \frac{11\pi^4}{45} g^6 + \dots \quad g^2 = \frac{g_{\text{YM}}^2 N_c}{16\pi^2}$$

$$\mathcal{Z}_{n+1} = \mathcal{Z}_n - \epsilon \mathcal{Z}_{n-1} + C \epsilon \tau \mathcal{Z}_1 + C' \epsilon^2 \mathcal{Z}_2$$

$\bar{Q}$ equation

Expanding for weak coupling:

$$\bar{Q} R_{n,k}^{1\text{-loop}} = \text{res}_{\epsilon=0} \int_{\tau=0}^{\tau=\infty} d^2|3 \mathcal{Z}_{n+1} \left(R_{n+1,k+1}^{\text{tree}} - R_{n,k}^{\text{tree}} R_{n+1,1}^{\text{tree}} \right) + \text{cyclic}$$

This relates Wilson loops to an integral over correlators of a lower loop order. The integration here is vastly simpler than standard loop integration and has allowed 2 and 3-loop amplitudes to be computed.

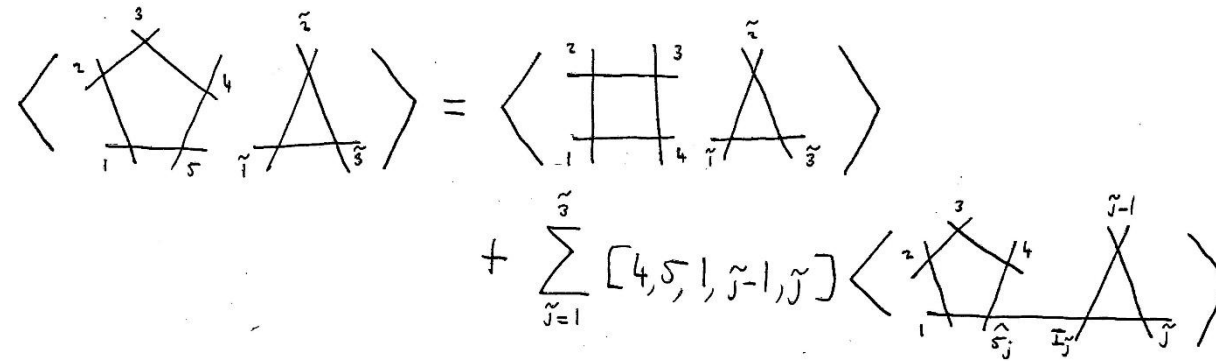
The $\bar{Q}$ equation can be generalised in a straight-forward manner to Wilson loop correlators [Drummond, MR, Wright, 2026], by putting the fermionic insertion on each Wilson loop.

$$\begin{aligned} \bar{Q}_A^{A'} R_{n_1,n_2}^{k,1\text{-loop}} &= \int \left[d^2|3 \mathcal{Z}_{X_1} \right]_A^{A'} \left(R_{n_1+1,n_2}^{k+1,\text{tree}} - R_{n_1,n_2}^{k,\text{tree}} R_{n_1+1}^{1,\text{tree}} \right) + \text{cyclic} \\ &+ \int \left[d^2|3 \mathcal{Z}_{X_2} \right]_A^{A'} \left(R_{n_1,n_2+1}^{k+1,\text{tree}} - R_{n_1,n_2}^{k,\text{tree}} R_{n_2+1}^{1,\text{tree}} \right) + \text{cyclic} \end{aligned}$$

As a first step towards using the $\bar{Q}$ equation, we verified it explicitly at 1-loop by making use of a recursion relation.

Wilson loop recursion

BCFW can be derived in the twistor Wilson loop picture, using the *loop equation*, which describes how a Wilson loop varies under deformations [Bullimore, Skinner, 2011]. The loop equation can be generalised in a natural way to multiple Wilson loops [Drummond, MR, Wright, to appear]. It gives, for example,



$$\langle \text{Pentagon}_{12345} \text{Triangle}_{\tilde{1}\tilde{2}\tilde{3}} \rangle = \langle \text{Square}_{1234} \text{Triangle}_{\tilde{1}\tilde{2}\tilde{3}} \rangle + \sum_{\tilde{j}=1}^{\tilde{3}} [4, 5, 1, \tilde{j}-1, \tilde{j}] \langle \text{Pentagon}_{1234\tilde{j}} \text{Triangle}_{\tilde{1}\tilde{2}\tilde{j}} \rangle$$

This gives us a form of the correlator with a smaller number of terms and a form of which allows the $\overline{Q}$ equation to be verified easily.

Summary

We introduced the formalism for computing Wilson loop correlators in twistor space in terms of Feynman diagrams. With this Feynman diagram expression, we were able to find that many of the interesting properties of scattering amplitudes are also satisfied by multiple Wilson loops:

BCFW-like Recursion relation:

$$\langle \text{Diagram 1} \rangle = \langle \text{Diagram 2} \rangle + \sum_{\tilde{j}=1}^{\tilde{j}} [4, \tilde{s}, \tilde{j}-1, \tilde{j}] \langle \text{Diagram 3} \rangle$$

Chiral box expansion and factorisation:

$$\text{Coefficient of } \langle \text{Diagram 1} \rangle = \langle \text{Diagram 2} \rangle + \langle \text{Diagram 3} \rangle + \langle \text{Diagram 4} \rangle + \langle \text{Diagram 5} \rangle$$

$\bar{Q}$ equation

$$\begin{aligned} \bar{Q}_A^{A'} R_{n_1, n_2}^{k, 1\text{-loop}} &= \int [d^{2|3} \mathcal{Z}_{X_1}]_A^{A'} \left(R_{n_1+1, n_2}^{k+1, \text{tree}} - R_{n_1, n_2}^{k, \text{tree}} R_{n_1+1}^{1, \text{tree}} \right) + \text{cyclic} \\ &+ \int [d^{2|3} \mathcal{Z}_{X_2}]_A^{A'} \left(R_{n_1, n_2+1}^{k+1, \text{tree}} - R_{n_1, n_2}^{k, \text{tree}} R_{n_2+1}^{1, \text{tree}} \right) + \text{cyclic} \end{aligned}$$

Ongoing work and future directions

Using these results, we are currently working on:

- Obtaining the 2-loop Wilson loop correlator by integrating the $\overline{Q}$ equation (and then the 3-loop correlator).
- Obtaining Wilson loop correlators from a light-like limit of a correlator of local operators. This could help us to get higher loop correlators.
- Find the geometry whose volume gives a Wilson loop correlator, analogous to the Amplituhedron for amplitudes.
- Factorisation and recursion for non-planar correlators.

Other possible directions:

- Reproducing $\langle W_{n_1} \mathcal{L} \rangle$ by taking limit of $\langle W_{n_1} W_{n_2} \rangle$. This is an important observable in $\mathcal{N} = 4$ SYM that could tell us about observables at arbitrary coupling and QCD amplitudes.
- Compute the correlator at strong coupling.
- Perform OPE on correlator to understand the spectrum of operators of $\mathcal{N} = 4$ SYM.

Thank you for listening!
Any questions?